

Higher Positive Coefficients in Integer Thue Morse Pressure

Research note prepared for Vladimir Reshetnikov with OpenAI

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Abstract

We prove that the coefficient immediately following the first positive post-cancellation coefficient in integer Thue–Morse pressure is also positive at every integer order $m \geq 2$. More generally, the coefficient of degree $2m + 2r$ is positive whenever $m \geq 3r$. The proof bounds higher derivatives of the normalized transfer eigenfunction by a summable inverse-branch series and isolates the two nearest half-integer branches. Exact rational calculations handle the few small orders outside the uniform bound. A negative degree-twelve coefficient at $m = 2$ shows that positivity does not persist at all subsequent degrees. This is a separate companion to the first-coefficient report.

Status and scope. The results are proved in ordinary mathematics and supported by exact finite certificates. This is an unrefereed research note, not a Lean formalization. The scope is integer pressure and its local Taylor coefficients. No uniform analytic remainder in a joint limit of phase and moment order, minimum sign-change degree in general, or worldwide priority claim is made.

1 Notation and results

For $T(x) = 2x \pmod{1}$, let $p_m(c)$ be the pressure of $m \log \cos^2 \pi(x - c)$ in the sum-over-preimages convention of [1]. The transfer operator is

$$(\mathcal{L}_{m,c}f)(x) = \sum_{Ty=x} \cos^{2m} \pi(y - c) f(y).$$

Near $c = 0$, it has a simple analytic eigenvalue $\rho_m(c) = \exp p_m(c)$ and eigenfunction $h_{m,c}$ normalized by $h_{m,c}(0) = 1$. Put

$$t = \pi c, \quad a = \tan t, \quad h_a = h_{m, \arctan(a)/\pi},$$

and define

$$R_m = \frac{2(2^{2m} - 1)\zeta(2m)}{\pi^{2m}}, \quad H_{m,r} = [a^{2r}]h_a(1/2), \quad E_{m,r} = [t^{2m+2r}]p_m(t/\pi).$$

Thus $H_{m,0} = R_m$. Write $B_m = H_{m,1}$ and $D_m = H_{m,2}$. The degree- $2m$ pressure coefficient vanishes [1], and the first-coefficient companion proves $B_m > 0$ and $E_{m,1} > 0$ for all $m \geq 2$ [2].

Theorem 1.1. *For every integer $m \geq 2$,*

$$D_m > 0, \quad E_{m,2} > 0.$$

More generally, for integers $r \geq 1$ and $m \geq 3r$,

$$H_{m,j} > 0 \quad (1 \leq j \leq r), \quad E_{m,r} > 0.$$

Together with the first-coefficient result, $E_{m,r} > 0$ holds for every $m \geq 2$ when $r = 1$ or $r = 2$.

The last conclusion does not extend to all m, r : Section 6 proves

$$E_{2,4} = [t^{12}]p_2(t/\pi) = -\frac{35360872}{93555} < 0.$$

Editorial note (ProveIt, 2026-10-01). Every positivity statement of Theorem 1.1 is contained in later packages of this series. For $m \geq 3r$ one has $j \leq r < m$, so $H_{m,j} > 0$ and $E_{m,r} > 0$ are cases of Theorem 1.1 of `../Thue_Morse_Integer_Pressure_Positive_Triangle/` ($1 \leq r < m$), which also gives $D_m > 0$ and $E_{m,2} > 0$ for $m \geq 3$. The remaining case $m = r = 2$, $D_2 = 536/27$ and $E_{2,2} = 6836/189$, is the case $m = 2$ of Theorem 1.1 of `../Thue_Morse_Integer_Pressure_Feedback_Boundary/` ($H_{m,m} > 0$ and $E_{m,m} > 0$ for every $m \geq 2$), and Theorem 1.1 of `../Thue_Morse_Integer_Pressure_Full_Range/` gives $E_{m,r} > 0$ for all $1 \leq r < 2m$. The two-nearest-branch argument of this note is a second route.

2 Higher inverse-branch coefficients

Normalize the operator by its fixed-point weight:

$$(V_a f)(x) = \sum_{Ty=x} (\cos \pi y + a \sin \pi y)^{2m} f(y).$$

Its eigenvalue satisfies the exact feedback equation

$$\lambda_m(a) = 1 + a^{2m} h_a(1/2), \quad p_m(t/\pi) = 2m \log \cos t + \log \lambda_m(\tan t). \quad (1)$$

Reflection makes $h_a(1/2)$ even in a .

The finite spectral inputs can be read directly on $\mathcal{E}_m = \text{span}\{e^{2\pi i k x} : 1 - m \leq k \leq m - 1\}$. The operator preserves this space, and its atomic eigenvalues are $1, 2^{-1}, \dots, 2^{-(2m-2)}$ $[1, 2]$. For example, the eigenfunctions

$$\sin^{2m}(\pi x) \sum_{n \in \mathbb{Z}} (\pi(x+n))^{-(2m-j)}, \quad 0 \leq j \leq 2m-2,$$

have eigenvalues 2^{-j} and distinct leading orders at zero. Analytic spectral projection therefore applies on a small complex disk in a .

For a half-integer w , define $G_{m,r}(w)$ by

$$G_{m,r}(w) = [a^{2r}] \prod_{j \geq 1} \left(1 + a \tan \frac{\pi w}{2^j}\right)^{2m}. \quad (2)$$

The product converges locally uniformly in a for each fixed w , because the tangent series is absolutely summable. No tangent has a pole: a scaled half-integer cannot be a half-integer modulo one.

Lemma 2.1. *For integers $r \geq 1$ and $m \geq r + 1$,*

$$H_{m,r} = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-2m} G_{m,r}(w), \quad (3)$$

with absolute convergence. Moreover,

$$H_{m,r} > \frac{2}{\pi^{2m}} \left[4^m \binom{2m}{2r} - \frac{(20m)^{2r}}{(2r)!} \left(\frac{2}{3}\right)^{2m-2r-2} \right]. \quad (4)$$

Proof. Let $F_N(a, x) = V_a^N 1(x)$. Evaluation at zero gives

$$F_{N+1}(a, 0) = F_N(a, 0) + a^{2m} F_N(a, 1/2),$$

so $F_N(a, 0) = 1 + O(a^{2m})$. Since $2r < 2m$, division by $F_N(a, 0)$ does not change its coefficient of degree $2r$ at $x = 1/2$. The normalized iterates converge uniformly analytically to h_a on a sufficiently small complex disk. Indeed, the leading projection of 1 is nonzero at zero and the other eigenvalues are uniformly smaller in modulus. Cauchy's formula yields

$$H_{m,r} = \lim_{N \rightarrow \infty} [a^{2r}] F_N(a, 1/2).$$

Represent the inverse branches by $W_N = \{w \in \mathbb{Z} + 1/2 : |w| < 2^{N-1}\}$. The cosine product telescopes, giving the exact finite identity

$$[a^{2r}] F_N(a, 1/2) = \sum_{w \in W_N} \left(2^N \sin \frac{\pi w}{2^N} \right)^{-2m} \cdot [a^{2r}] \prod_{j=1}^N \left(1 + a \tan \frac{\pi w}{2^j} \right)^{2m}. \quad (5)$$

For every half-integer w , the elementary dyadic bound from [2] is

$$\sum_{j \geq 1} \left| \tan \frac{\pi w}{2^j} \right| < 10|w|. \quad (6)$$

For completeness, take $w > 0$ and $J = \lceil \log_2(2w) \rceil$. For $j \leq J$, distance from a tangent pole gives $|\tan(\pi w/2^j)| < 2^{j+1}/\pi$. For $j \geq J+1$, the argument is at most $\pi/4$, so convexity gives $|\tan(\pi w/2^j)| \leq 4w/2^j$. Summing yields a bound below $(16/\pi + 4)w < 10w$.

The absolute degree- $2r$ coefficient of the finite product is bounded by the elementary symmetric sum of $2m$ copies of every tangent magnitude, hence by

$$\frac{1}{(2r)!} \left(2m \sum_{j \geq 1} \left| \tan \frac{\pi w}{2^j} \right| \right)^{2r} < \frac{(20m|w|)^{2r}}{(2r)!}.$$

On W_N , the unperturbed factor in (5) is at most $(2|w|)^{-2m}$. Thus the absolute summand is bounded by $O_{m,r}(|w|^{2r-2m})$, summable for $m \geq r+1$. Dominated convergence proves (3).

At $w = 1/2$ all tangent terms are positive, and the first is 1. Choosing $2r$ of its $2m$ copies gives $G_{m,r}(1/2) \geq \binom{2m}{2r}$; the even coefficient at $w = -1/2$ is equal. For every other branch use the absolute bound. Strict convexity gives

$$\sum_{k \geq 1} (k + 1/2)^{-2} < \int_1^\infty x^{-2} dx = 1,$$

and therefore

$$\sum_{k \geq 1} (k + 1/2)^{-(2m-2r)} < \left(\frac{2}{3} \right)^{2m-2r-2}.$$

Combining the two nearest branches with the remaining absolute tail proves (4). $\square$

3 The fourth response and the next pressure coefficient

At $r = 2$, the bracket in (4) is

$$U_m = 4^m \binom{2m}{4} - \frac{(20m)^4}{24} \left(\frac{2}{3}\right)^{2m-6}.$$

Its value at $m = 6$ is $34263040/27 > 0$. The first term increases with m , while the ratio of successive second terms is at most

$$\left(\frac{7}{6}\right)^4 \frac{4}{9} = \frac{2401}{2916} < 1 \quad (m \geq 6).$$

Thus $D_m > 0$ for every $m \geq 6$. The remaining exact response values are

$$D_2 = \frac{536}{27}, \quad D_3 = \frac{16454}{315}, \quad D_4 = \frac{136433288}{1845585}, \quad D_5 = \frac{54583453322}{703167885}. \quad (7)$$

An exact finite-matrix certificate for these rational values is described in Section 5. In particular, the inverse-branch derivative argument is not applied at $(m, r) = (2, 2)$, where its summability hypothesis fails.

Expanding (1) gives, for $m \geq 3$,

$$E_{m,2} = D_m + \frac{2m+2}{3}B_m + \frac{m(10m+7)}{45}R_m - \frac{m}{m+2}R_{m+2}. \quad (8)$$

The relative degree-four coefficient of $\tan^{2m} t$ is $m(10m+7)/45$, and the logarithm's square starts at degree $4m > 2m+4$. Since

$$R_{m+2} \leq \left(\frac{4}{\pi^2}\right)^2 R_m,$$

the last two terms of (8) have positive sum. Indeed, their bracket is at least

$$mR_m \left(\frac{10m+7}{45} - \frac{16}{(m+2)\pi^4} \right) > 0.$$

Together with $B_m, D_m > 0$, this proves $E_{m,2} > 0$ for $m \geq 3$. For $m = 2$, (8) has the additional logarithmic correction $-R_2^2/2$ and gives

$$E_{2,2} = \frac{6836}{189} > 0.$$

4 A uniform region of positive coefficients

Proposition 4.1. *For $r \geq 3$ and $m \geq 3r$, the bracket in (4) is strictly positive.*

Proof. The ratio of its positive term to its subtracted term is

$$\frac{4^m (2m)_{2r}}{(20m)^{2r}} \left(\frac{3}{2}\right)^{2m-2r-2},$$

where $(2m)_{2r}$ is a falling factorial. For $m \geq 3r$,

$$(2m)_{2r} \geq (2m)^{2r} \left(1 - \frac{r}{m}\right)^{2r} \geq (2m)^{2r} \left(\frac{2}{3}\right)^{2r}.$$

The ratio is consequently at least

$$\frac{4}{9} 9^m \left(\frac{4}{2025} \right)^r \geq \frac{4}{9} \left(\frac{36}{25} \right)^r > 1.$$

The last inequality holds at $r = 3$, when its value is $20736/15625 > 1$, and then increases with r . $\square$

If $m \geq 3r$, Proposition 4.1, the fourth-response result, and $B_m > 0$ imply $H_{m,j} > 0$ for every $1 \leq j \leq r$. Because $r < m$, the logarithm's square in (1) starts beyond degree $2m + 2r$. Hence

$$E_{m,r} = \sum_{j=0}^r H_{m,j} [t^{2m+2r}] \tan^{2m+2j} t - \frac{m}{m+r} R_{m+r}. \quad (9)$$

All tangent coefficients in the sum are nonnegative. The $j = 0$ term alone dominates the negative term: since $\tan t/t = 1 + t^2/3 + \dots$ has positive coefficients,

$$\begin{aligned} [t^{2r}] (\tan t/t)^{2m} &\geq \frac{\binom{2m}{r}}{3^r} \geq \left(\frac{2m}{3r} \right)^r \geq 2^r, \\ \frac{m}{m+r} \frac{R_{m+r}}{R_m} &\leq \frac{m}{m+r} \left(\frac{4}{\pi^2} \right)^r < 1. \end{aligned}$$

The binomial bound follows by multiplying $(2m-j)/(r-j) \geq 2m/r$ for $0 \leq j < r$. Positivity of the tangent coefficients also follows directly from $(\tan t)' = 1 + \tan^2 t$. This proves the remaining conclusions of Theorem 1.1.

5 Exact finite certificates

The small response values use the Fourier basis with indices $1 - m, \dots, m - 1$. If $j = 2k - \ell$, the corresponding entry of V_a is

$$2^m \binom{2m}{m+j} (1 - ia)^{m+j} (1 + ia)^{m-j}, \quad (10)$$

with value zero when $|j| > m$. Write $V_a = \sum_n i^n M_n a^n$, $h_a = \sum_n i^n v_n a^n$, and $\lambda(a) = \sum_n i^n \ell_n a^n$. The M_n are rational matrices, explicitly

$$(M_n)_{k,\ell} = (M_0)_{k,\ell} \sum_{q=0}^n (-1)^q \binom{m+j}{q} \binom{m-j}{n-q}.$$

Normalize $\mathbf{1}^T v_0 = 1$ and $\mathbf{1}^T v_n = 0$ for $n > 0$. Given earlier coefficients, put

$$b_n = \sum_{q=1}^n M_q v_{n-q}, \quad \ell_n = \mathbf{1}^T b_n,$$

and solve

$$(I - M_0) v_n = b_n - \sum_{q=1}^n \ell_q v_{n-q}, \quad \mathbf{1}^T v_n = 0. \quad (11)$$

Evaluation at $x = 1/2$ means summing Fourier coefficients with signs $(-1)^k$. Thus $D_m = \sum_k (-1)^k (v_4)_k$. The exact checker verifies every residual and produces (7). It also compares the resulting $E_{m,2}$ values with an independent direct pressure recursion through $m = 8$; additional response values through $m = 12$ are supplied as regression checks.

6 A negative later coefficient

The source gives the order-two characteristic polynomial

$$\chi_2(\lambda, C) = \lambda^3 - (3/4 + C)\lambda^2 + (1/4 + 5C/8)\lambda - 1/8, \quad C = \cos 2t.$$

Its atomic branch has $\lambda(0) = 1$ and $\partial_\lambda \chi_2(1, 1) = 3/8 \neq 0$. Direct rational series substitution gives

$$\begin{aligned} p_2(t/\pi) = & -2t^2 + \frac{376}{45}t^6 + \frac{6836}{189}t^8 + \frac{105448}{2025}t^{10} \\ & - \frac{35360872}{93555}t^{12} + O(t^{14}). \end{aligned}$$

The verifier substitutes the exponential of this series into χ_2 , together with the cosine series, and checks an identically zero residual through degree twelve. The simple-root derivative makes those Taylor coefficients unique. In particular $E_{2,4} < 0$, so no claim of positivity at every subsequent even degree is possible.

7 Provenance and remaining questions

The source manuscript [1] proves the missing coefficient and asks about the first following coefficient. The separate companion [2] proves positivity at that first degree. This note extends the argument to the next degree and to the stated uniform region; it does not revise the earlier artifacts. The bounded exact calculations in the package are not substitutes for the general estimates.

The region $m \geq 3r$ is sufficient, not asserted optimal. The signs outside it, the first negative degree at each fixed moment order, and the possibility of a sharper uniform boundary remain open here. A coefficient estimate does not supply a uniform radius of phase analyticity as m varies.

Editorial note (ProveIt, 2026-10-01). The questions of this paragraph other than the uniform radius are answered by later packages of the series. Outside the region $m \geq 3r$ the pressure coefficients are still positive up to degree $6m - 2$: $E_{m,r} > 0$ for all $1 \leq r < 2m$ (`../Thue_Morse_Integer_Pressure_Full_Range/`), and $E_{2,4} < 0$ shows that degree $6m$ cannot be included in general. The responses satisfy $H_{m,r} > 0$ for $1 \leq r \leq m$ (`../Thue_Morse_Integer_Pressure_Positive_Triangle/`, `../Thue_Morse_Integer_Pressure_Feedback_Boundary/`), while $H_{2,3} = -488/27 < 0$. The first negative degree is treated in `../Thue_Morse_Integer_Pressure_Sign_Changes/`, `../Thue_Morse_Integer_Pressure_First_Negative/` and `../Thue_Morse_Integer_Pressure_First_Negative_Data/`.

Editorial note (ProveIt, 2026-10-01). This note is the second of thirteen packages written in one series and filed together on 2026-10-01 beside the repository manuscript `../Thue_Morse_Integer_Pressure/` in the Fabius drafts tree (batch 72 of the repository's `docs/incoming/`; unreviewed). In logical order they are the positivity series `../Thue_Morse_Integer_Pressure_First_Positive/`, `../Thue_Morse_Integer_Pressure_Higher_Positive/`, `../Thue_Morse_Integer_Pressure_Positive_Triangle/`, `../Thue_Morse_Integer_Pressure_Feedback_Boundary/`, `../Thue_Morse_Integer_Pressure_Beyond_Boundary/`, `../Thue_Morse_Integer_Pressure_Linear_Region/` and `../Thue_Morse_Integer_Pressure_Full_Range/`, and the sign series, which imports the full-range theorem: `../Thue_Morse_Integer_Pressure_Sign_Changes/`, `../Thue_Morse_Integer_Pressure_Sign_Densities/`, `../Thue_Morse_Integer_Pressure_Negative_Bound/`, `../Thue_Morse_Integer_Pressure_First_Negative/` and `../Thue_Morse_Integer_Pressure_Cluster_Asymptotics/`,

with the dataset `../Thue_Morse_Integer_Pressure_First_Negative_Data/` (no article). An earlier version of `../Thue_Morse_Integer_Pressure_Full_Range/`, *The Full Pressure Positivity Range for Large Integer Orders* (orders $m \geq 4096$), delivered in the same batch, is superseded by it and was not filed. Here [1] is `../Thue_Morse_Integer_Pressure/` (the version cited differs from the filed manuscript only by dated editorial changes, none of them mathematical), and [2] is `../Thue_Morse_Integer_Pressure_First_Positive/` (main source `positive_thue_morse_coefficient.tex`).

References

- [1] *Integer Pressure and a Missing Taylor Coefficient*. ProveIt research manuscript, 28 September 2026, current version inspected 1 October 2026. Source commit `6a98f89bac85cc6b4ba5d5c3b63996037d9824e1`, blob `1973fd17245864bb8a83e1b1d90416ac032ba7c1`. Versioned repository source.
- [2] *The First Positive Coefficient in Integer Thue Morse Pressure*. Research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. Separate companion with finite Bernoulli and inverse-branch proofs, independently checked exact calculations, and an explicit large-order expansion.